4D Scene Reconstruction from Monocular Video using Gaussian Splatting

Ahmad Rashad
Student ID: XXI-XXXX
Department of Computer Science

National University of Computer and Emerging Sciences
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Chapter 1

Introduction

Recent developments in computer vision and computer graphics have enabled machines to reconstruct three-dimensional environments from ordinary images and videos. Techniques such as Neural Radiance Fields (NeRF) and Gaussian Splatting have shown promising results in generating high-quality scene representations that allow rendering from previously unseen viewpoints.

These technologies are increasingly used in fields such as virtual reality, robotics, autonomous navigation, digital heritage preservation, and immersive media production.

This project investigates the reconstruction of dynamic scenes from monocular video using Gaussian Splatting. The goal is to explore whether a single video captured using a conventional camera can be used to reconstruct a navigable three-dimensional scene representation.

Unlike traditional 3D reconstruction systems that rely on stereo cameras or specialized sensors such as LiDAR, this research focuses on reconstructing scene geometry using only monocular video data. Achieving this is significantly more challenging because depth information must be inferred indirectly from motion and visual features present in the frames.

The project therefore aims to design and implement a pipeline that extracts frames from a video, estimates camera poses, reconstructs scene geometry, and generates a view-synthesized representation of the scene using Gaussian Splatting techniques.

1.1 Problem Domain

3D scene reconstruction has traditionally relied on specialized hardware such as stereo cameras, structured light sensors, or LiDAR systems. While these approaches produce accurate depth information, they are expensive and often impractical for general users.

Recent research has demonstrated that neural rendering methods can reconstruct scenes from ordinary images or videos using machine learning techniques. Methods such as Neural Radiance Fields (NeRF) and Gaussian Splatting learn a representation of the scene that allows rendering from new viewpoints.

However, reconstructing a reliable 3D or 4D representation from a single monocular video remains a difficult problem. Several practical challenges arise during the reconstruction process:

- Obtaining accurate camera pose estimation from monocular video frames
- Ensuring sufficient scene coverage for reconstruction
- Handling motion blur and redundant frames
- Dealing with scenes that lack distinctive visual features
- Achieving stable Gaussian Splatting optimization
- Managing high computational requirements during training

Another major challenge is extending the reconstruction from a static 3D representation to a dynamic 4D representation where the scene evolves over time. Modeling temporal changes accurately requires more complex algorithms and representations.

Due to these challenges, the reconstruction pipeline requires extensive experimentation, preprocessing, and parameter tuning in order to produce visually meaningful results.

1.2 Research Problem Statement

The primary objective of this research is:

To design and implement a system that reconstructs a 3D representation of a scene from a monocular video using Gaussian Splatting and enables rendering from novel viewpoints.

The research attempts to answer the following questions:

- How accurately can a 3D scene be reconstructed from a single monocular video?
- What preprocessing steps improve reconstruction quality?
- How effective is Gaussian Splatting compared to traditional NeRF-based approaches?

1.3 Software Requirements Specification

1.3.1 Functional Requirements

- Accept a monocular video as input
- Extract frames from the input video
- Perform camera pose estimation using COLMAP
- Train a Gaussian Splatting model
- Render novel viewpoints of the reconstructed scene
- Generate a 3D visualization of the reconstructed environment

1.3.2 Non-Functional Requirements

- High reconstruction quality
- Efficient training performance
- Real-time or near real-time rendering
- Scalability for larger scenes

1.3.3 Tools and Technologies

- Python
- PyTorch
- COLMAP
- Gaussian Splatting framework
- Google Colab / GPU computing
- MeshLab for point cloud visualization

Chapter 2

Literature Review

This chapter reviews existing research in the field of neural rendering and 3D reconstruction in order to understand current approaches and their limitations.

2.1 Related Research

2.1.1 Research Item 1: Neural Radiance Fields (NeRF)

Summary

Neural Radiance Fields (NeRF) introduced a method for synthesizing novel views of complex scenes by representing the scene as a continuous volumetric function. The model uses a neural network to predict color and density for every 3D coordinate and viewing direction.

NeRF demonstrated impressive visual quality and became a milestone in neural rendering research.

Critical Analysis

Although NeRF produces high-quality renderings, it suffers from slow rendering speeds because each pixel requires evaluating a neural network multiple times along a viewing ray.

Training NeRF models also requires significant GPU resources.

Relationship to Proposed Research

Our work builds upon the concept of neural scene representation but replaces volumetric neural networks with Gaussian primitives in order to achieve faster rendering.

2.1.2 Research Item 2: Instant Neural Graphics Primitives (Instant-NGP)

Summary

Instant-NGP accelerates NeRF training using multi-resolution hash encoding. It significantly reduces training time and enables faster rendering of neural scenes.

Critical Analysis

Although training speed improves significantly, neural network inference is still required during rendering, which may limit performance.

Relationship

This work demonstrates the importance of efficient scene representation, which is further improved by Gaussian Splatting.

2.1.3 Research Item 3: 3D Gaussian Splatting

Summary

Gaussian Splatting represents scenes using a collection of 3D Gaussians. Each Gaussian stores position, color, opacity, and covariance parameters. Rendering is performed by projecting these Gaussians onto the screen.

Critical Analysis

The technique enables real-time rendering and high-quality scene reconstruction. However, the quality of results depends heavily on accurate camera pose estimation.

Relationship

This method forms the foundation of the proposed research.

2.1.4 Research Item 4: Structure-from-Motion using COLMAP

Summary

COLMAP is a widely used tool for performing structure-from-motion reconstruction and estimating camera poses from image collections.

Critical Analysis

COLMAP performs reliably when images contain sufficient visual features but may struggle with motion blur or low-texture scenes.

Relationship

COLMAP is used in our preprocessing pipeline to estimate camera parameters.

2.1.5 Research Item 5: Dynamic Scene Reconstruction

Summary

Dynamic reconstruction techniques extend neural rendering models to capture time-varying scenes.

Critical Analysis

These methods require significantly higher computational cost and more complex models.

Relationship

Our project aims to explore whether Gaussian Splatting can eventually support dynamic scene reconstruction.

2.2 Analysis Summary

Method	Strength	Weakness	Relevance
NeRF	High rendering quality	Slow rendering	Baseline method
Instant-NGP	Fast training	Neural inference required	Optimization inspira
Gaussian Splatting	Real-time rendering	Sensitive to camera poses	Core technique
COLMAP	Accurate pose estimation	Requires good features	Preprocessing step
Dynamic NeRF	Handles motion	High complexity	Future extension

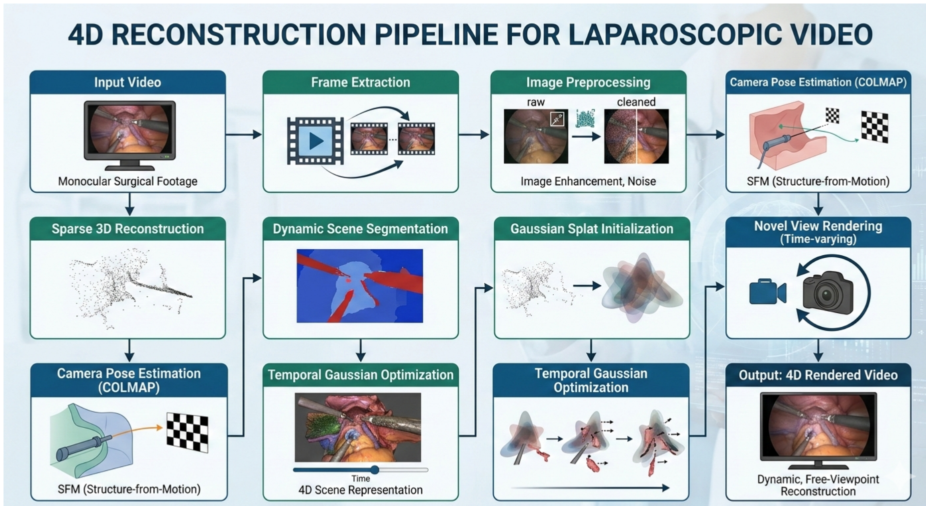
Chapter 3

Proposed Approach

The proposed system reconstructs a 3D scene from a monocular video using a multi-stage pipeline.

3.1 System Pipeline

1. Video capture
2. Frame extraction
3. Camera pose estimation using COLMAP
4. Sparse point cloud generation
5. Gaussian Splatting training
6. Novel view rendering



Architecture pipeline of the proposed 4D reconstruction system. The pipeline processes input video frames, estimates camera poses using COLMAP, trains a Gaussian Splatting model, and renders novel views to generate a 4D scene representation.

3.2 Frame Extraction

The input video is converted into individual frames. These frames serve as the dataset for the reconstruction pipeline.

3.3 Camera Pose Estimation

COLMAP is used to estimate the camera position and orientation for each frame. This step also produces a sparse point cloud that represents the scene geometry.

3.4 Gaussian Splatting Training

The Gaussian Splatting model optimizes a collection of 3D Gaussian primitives based on the input images and camera parameters.

3.5 Novel View Rendering

After training, the scene can be rendered from new viewpoints by projecting the Gaussian primitives onto the image plane.

3.6 Current Progress and Challenges

During the implementation of the reconstruction pipeline, several practical challenges have been encountered.

The first challenge involved preparing the input dataset. Extracting frames from the video and ensuring sufficient scene coverage required experimentation with different frame sampling strategies.

Another major difficulty was achieving stable camera pose estimation using COLMAP. Monocular videos often contain redundant viewpoints and motion blur, which can reduce the number of reliable feature correspondences required for accurate reconstruction.

After generating the sparse point cloud, the Gaussian Splatting training process was executed. Although the training pipeline completed successfully, the resulting point cloud sometimes appeared noisy or blob-like, indicating that the reconstruction parameters require further tuning.

Additionally, reconstructing a dynamic 4D scene from a monocular video introduces further complexity, as the system must account for temporal variations in geometry and appearance.

At the current stage of the project, the preprocessing pipeline and training workflow have been successfully implemented. However, achieving a stable and visually accurate reconstruction remains an ongoing process.

The research team is continuously experimenting with improved preprocessing methods, frame selection strategies, and optimized training parameters in order to obtain a more accurate scene reconstruction.

Although a complete 4D reconstruction has not yet been fully achieved, continuous progress is being made toward producing an optimal and acceptable reconstruction result.

3.7 Expected Outcome

The expected outcome of this project is a system capable of reconstructing a scene representation from a monocular video.

The final system is expected to produce:

- A reconstructed 3D point cloud of the scene
- Rendered images from novel viewpoints
- A synthesized video demonstrating the reconstructed scene
- Visualizations of Gaussian primitives and reconstructed geometry

The primary objective is to achieve a visually consistent reconstruction that demonstrates the feasibility of 3D scene reconstruction from monocular video data.

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